COMPUTER VISION

INTRODUCTION

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Contents

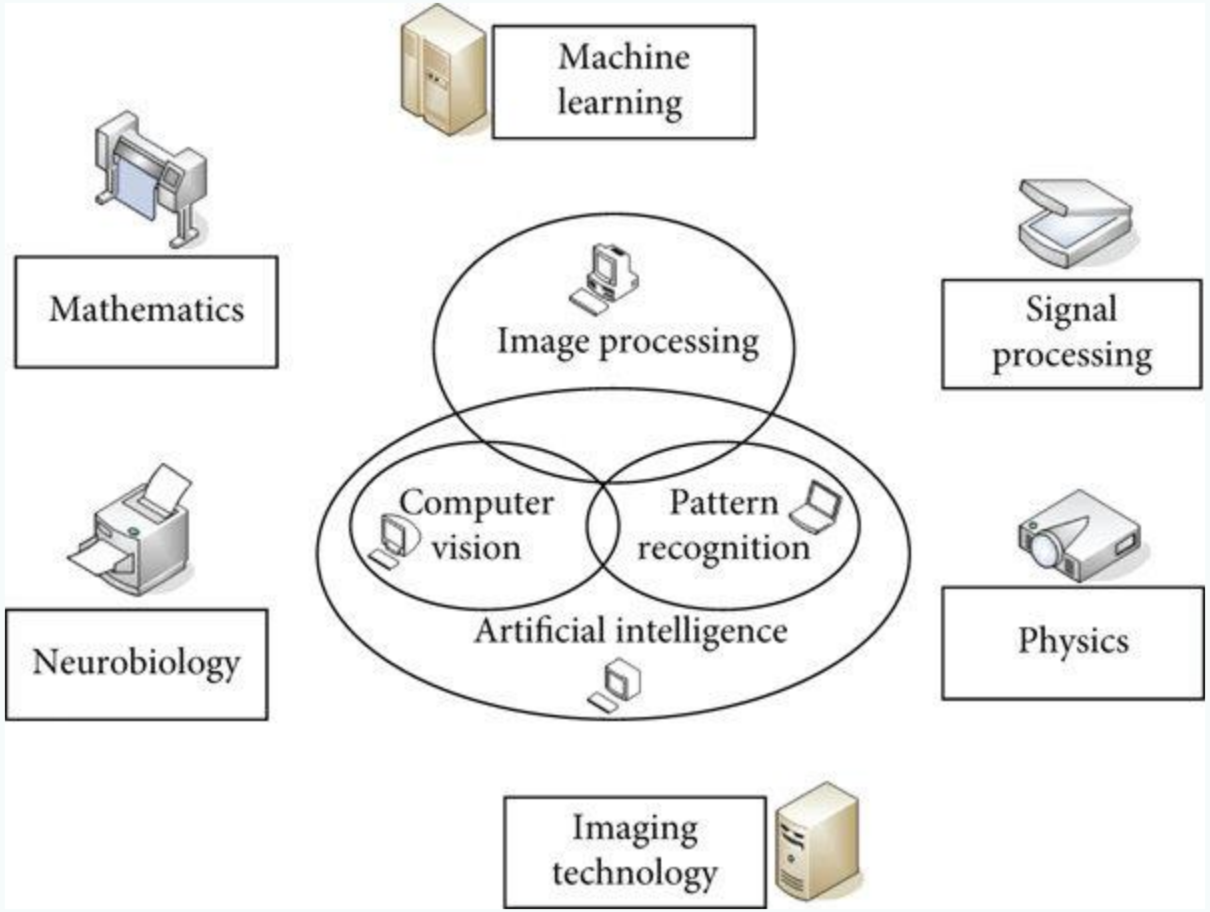
Lecture outcomes

- Understanding what are computer vision, some its applications and main challenges.
- Implementing the getting started tiny application in CV by loading, showing image/video and putting your brand (text or icon) on the frames.

What is computer vision

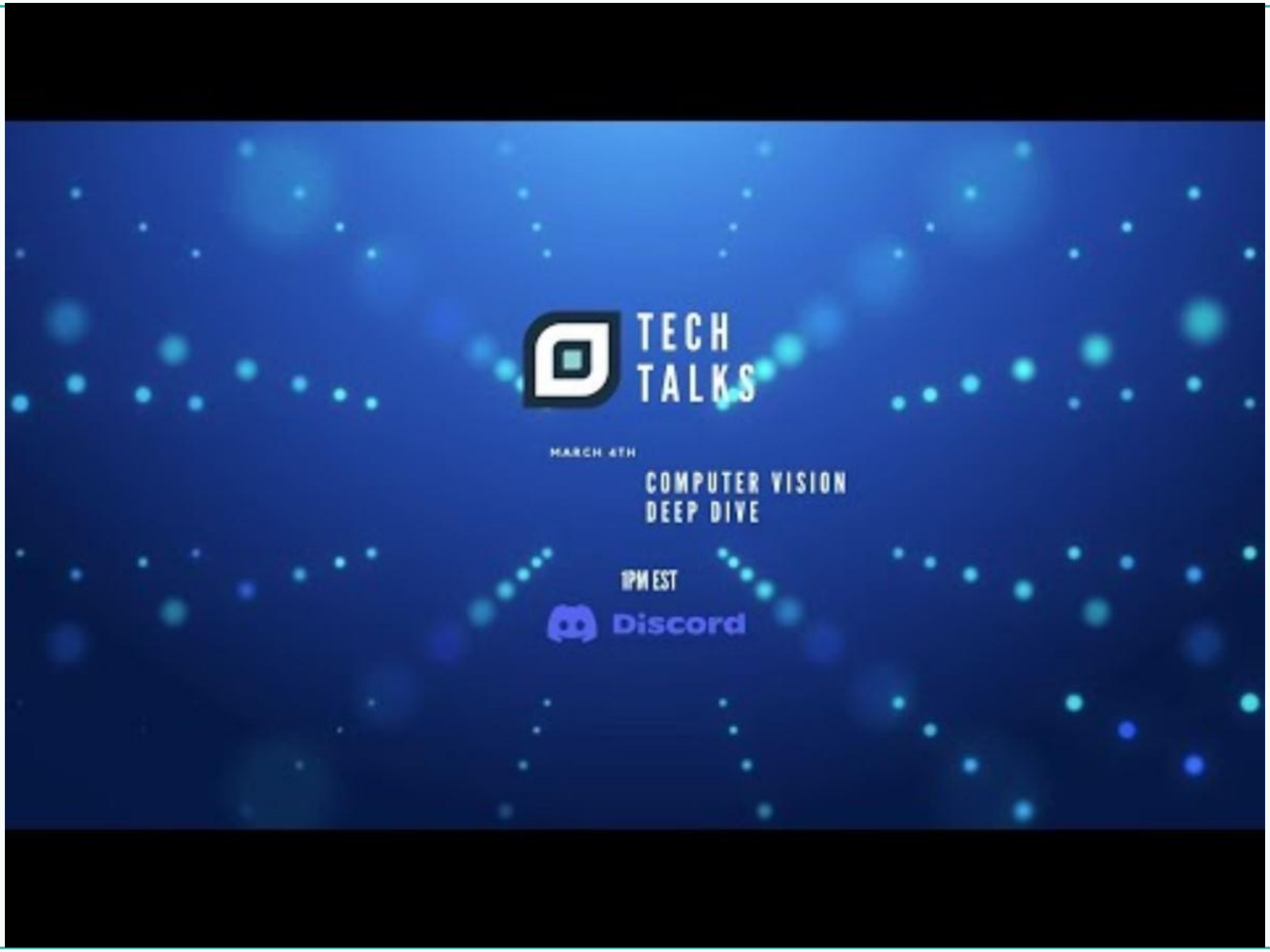
- Computer vision is a field of artificial intelligence (AI) that enables computers and systems to derive meaningful information from digital images, videos and other visual inputs-and take actions or make recommendations based on that information(https://www.ibm.com/topics/computer-vision).
- Computer vision is a field of computer science that focuses on enabling computers to identify and understand objects and people in images and videos. Like other types of AI, computer vision seeks to perform and automate tasks that replicate human capabilities. In this case, computer vision seeks to replicate both the way humans see, and the way humans make sense of what they see (https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-is-computer-vision#object-classification).
- Computer vision, a fascinating field at the intersection of computer science and artificial intelligence, which enables computers to analyze images or video data, unlocking a multitude of applications across industries, from autonomous vehicles to facial recognition systems (https://www.geeksforgeeks.org/computer-vision/).

Computer vision and related disciplines



https://www.researchgate.net/figure/Relationship-between-computer-vision-and-related-disciplines_fig2_360146441

Brief history of computer vision



https://youtu.be/qHvsocj1c08

Brief history of computer vision

- 1960s: Researchers aimed to enable computers to interpret visual information. MIT AI lab begins computer vision projects.
- 1970s: Optical Character Recognition (OCR) debuts via Kurzweil Computer Products.
- 1980s: Smart cameras, such as the Xerox optical mouse, and layered image processing.
- 2000s: Viola-Jones object detection framework (facial recognition and object detection); Pascal VOC Project launched for developing image datasets for training.

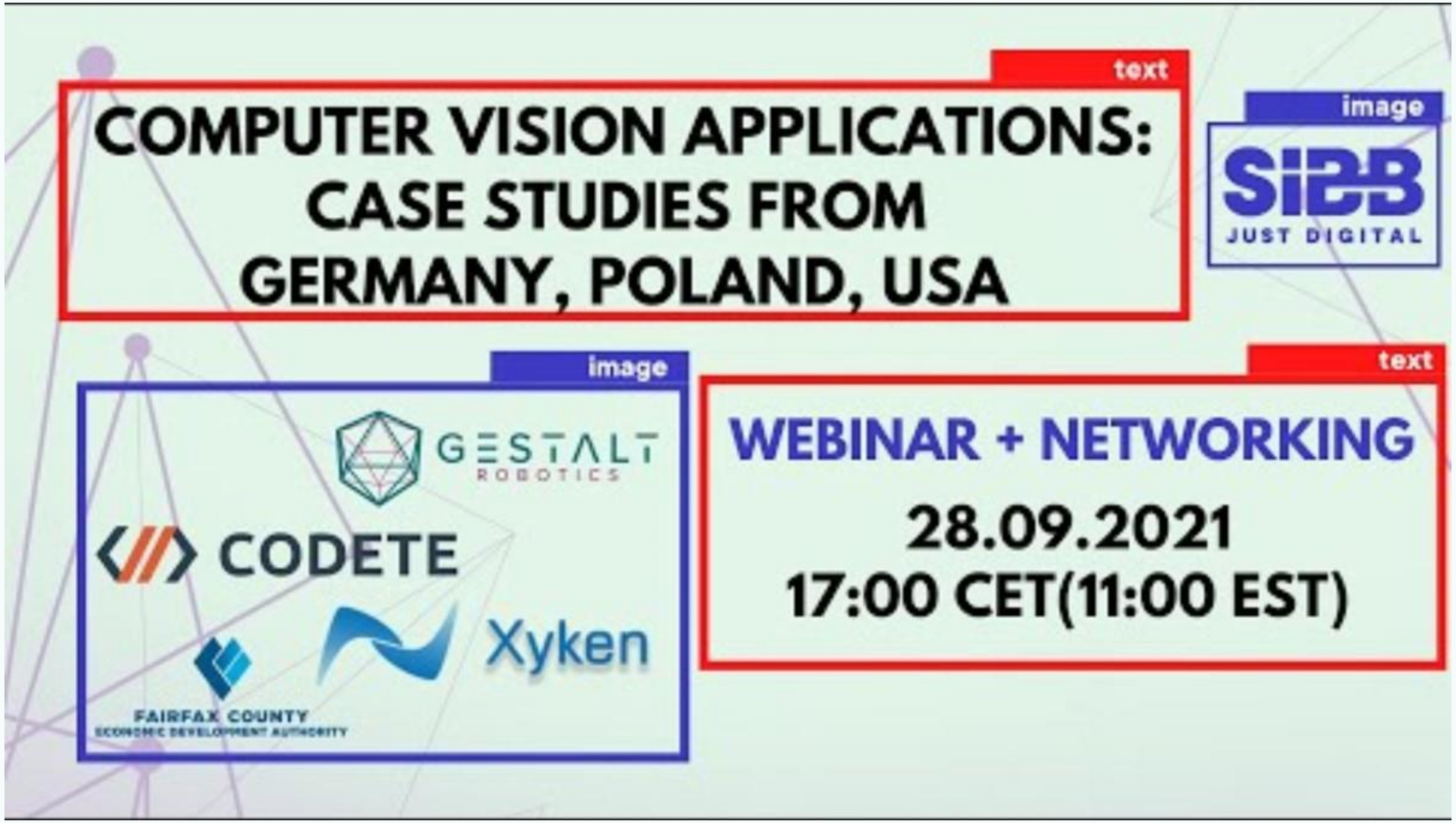
Brief history of computer vision

- 2005: Security Show Japan 2005 and the first mobile phone with working facial recognition features.
- 2010: AI's average object-recognition error rate becomes better than humans'; 2.5% versus 5%; First ImageNet competition. Developers competed for developing image recognition models. ImageNet now has up to 15 million images in 20,000 categories.
- 2015: Google introduces FaceNet for facial recognition. Requires minimum data set.
- 2018: NVIDIA announces that it will make its hyper-realistic face generator StyleGAN open source.

Some applications of computer vision

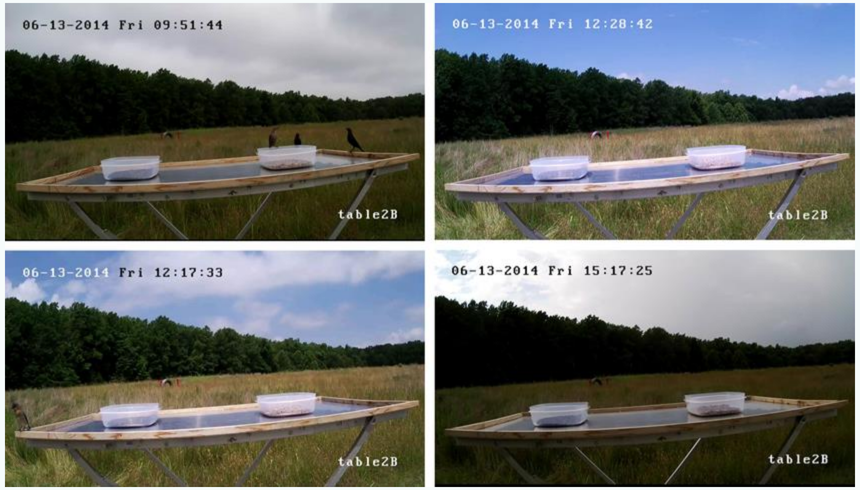
Some applications of computer vision

Some applications of computer vision



The challenges in computer vision

- **Variations in Lighting**: Changes in lighting conditions can drastically alter the appearance of an object, making it difficult for computer vision systems to consistently recognize objects.
- **Changes in Perspective**: While humans are generally good at recognizing objects from different perspectives, this is a significant challenge for computer vision systems.
- **Scale Invariance**: Another challenge is recognizing objects of different sizes or when the size of the object changes due to changes in distance



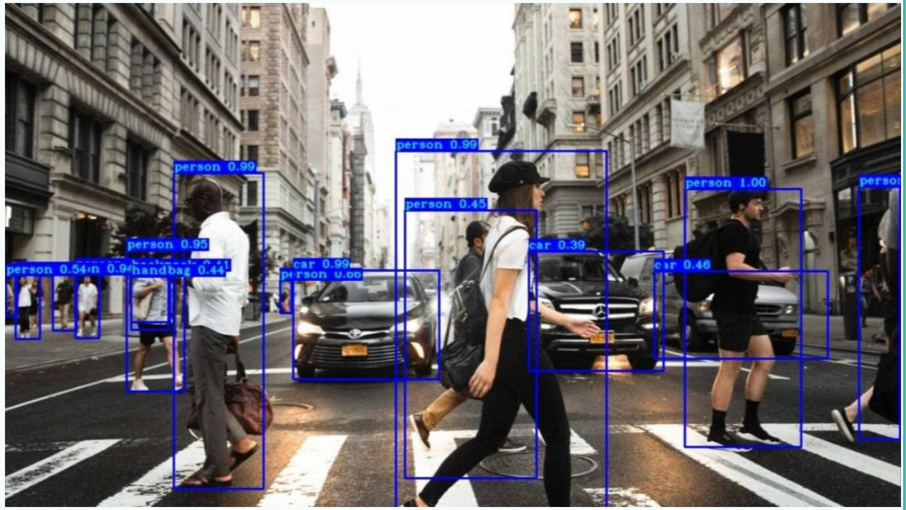
Some of the lighting variations that occurred in one day of testing. Changes in lighting added complexity for computer vision due to shadows and reflections on the food tables

The challenges in computer vision

- **Occlusion**: Occlusion occurs when an object of interest is partially hidden by other objects. Recognizing an object when only part of it is visible is a complex problem

- **Scene Complexity**: Real-world scenes can be incredibly complex, with many objects interacting in various ways. Understanding these scenes, especially when the objects are similar or overlapping.

- **Real-Time Processing**: Many applications, such as autonomous driving or video surveillance, require real-time processing



handling complex scenes with occlusions and cluttered backgrounds, scale and perspective variations, object class imbalance, and achieving real-time performance on resource-constrained devices

Getting started with tiny application

- Capture video or read MP4 from device
- Override your name/icon at the left/right top corner of the frames
- Convert the frames to gray space